

COUNCIL BLUFFS, IA, USA

WMO: 725497

Lat: 41.259N

Lon: 95.760W

Elev: 382

StdP: 96.82

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-16.3	-23.7	0.5	-17.9	-21.3	0.6	-14.7	12.4	-3.7	11.3	-4.4	3.8	330	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	33.9	25.5	32.5	24.9	31.2	24.1	27.7	31.8	26.3	30.6	25.3	29.5	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	23.7	31.1	25.1	21.1	29.2	23.9	19.6	28.1	90.7	32.0	84.5	30.8	79.3	29.4	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.6	8.6	-22.7	36.3	2.8	1.2	-24.7	37.2	-26.3	37.8	-27.8	38.5	-29.8	39.4		
(5)			-22.2	29.0	3.2	1.3	-24.5	30.0	-26.4	30.8	-28.2	31.5	-30.5	32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-3.9	-1.8	4.7	11.4	17.3	22.9	24.4	23.6	19.6	12.4	4.5	-1.8	(6)
(7)		DBStd	11.45	6.74	6.86	6.88	5.58	4.86	3.48	3.25	3.11	4.41	5.19	5.95	6.60	(7)
(8)		HDD10.0	1593	429	332	192	49	4	0	0	0	1	34	181	370	(8)
(9)		HDD18.3	3251	688	563	424	217	78	5	1	2	38	192	415	627	(9)
(10)		CDD10.0	2022	1	2	28	90	231	386	447	421	288	109	17	3	(10)
(11)		CDD18.3	637	0	0	2	8	47	142	189	165	74	8	1	1	(11)
(12)		CDH23.3	5901	0	0	16	99	436	1323	1858	1416	656	78	4	16	(12)
(13)		CDH26.7	2174	0	0	2	28	142	494	734	532	221	16	0	7	(13)
(14)	Wind	WSAvg	4.1	4.3	4.4	4.8	5.0	4.4	3.8	3.1	3.0	3.6	4.1	4.3	4.1	(14)
(15)	Precipitation	PrecAvg	842	23	28	52	86	124	131	100	120	63	53	33	26	(15)
(16)		PrecMax	1096	43	64	114	188	234	286	244	262	169	132	69	122	(16)
(17)		PrecMin	567	8	7	6	12	43	18	8	32	10	0	0	0	(17)
(18)		PrecStd	142	10	14	28	37	53	68	56	66	43	37	22	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	18.0	25.2	30.2	33.6	34.9	36.2	35.8	33.0	28.8	22.8	23.8	(19)
(20)			MCWB	7.3	11.4	15.0	18.3	22.2	24.0	27.0	25.7	23.7	19.1	14.6	18.2	(20)
(21)		2%	DB	10.2	13.9	22.0	26.1	30.1	32.8	33.8	32.9	32.0	25.9	18.9	12.4	(21)
(22)			MCWB	5.7	8.9	14.3	16.1	21.0	24.2	27.3	25.8	23.5	17.3	13.6	8.3	(22)
(23)		5%	DB	7.4	10.9	17.9	22.8	27.7	31.3	32.4	31.4	29.0	22.9	16.1	9.1	(23)
(24)			MCWB	4.1	7.0	12.3	14.9	19.6	23.5	26.1	24.8	22.3	15.9	10.6	6.0	(24)
(25)		10%	DB	5.0	7.5	15.1	20.2	25.9	29.8	30.9	29.8	27.3	21.1	12.9	6.5	(25)
(26)			MCWB	2.4	4.4	10.1	13.6	18.4	22.6	25.2	24.2	20.8	15.1	8.6	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	13.4	17.9	19.7	24.2	27.6	30.0	28.8	26.0	21.7	16.8	18.4	(27)
(28)			MCD B	12.8	15.3	22.5	24.6	29.1	30.7	34.1	33.3	31.1	26.1	20.5	23.5	(28)
(29)		2%	WB	6.0	9.4	15.5	17.9	22.6	25.8	28.3	26.9	24.5	19.3	14.0	9.0	(29)
(30)			MCD B	9.9	13.7	19.7	24.2	27.9	30.4	32.3	31.2	29.2	23.3	17.6	11.1	(30)
(31)		5%	WB	4.2	6.8	13.1	16.2	21.2	24.7	26.8	25.8	23.4	17.6	11.6	6.0	(31)
(32)			MCD B	6.9	10.5	17.5	21.6	25.9	29.6	31.0	30.0	27.7	20.9	15.9	8.9	(32)
(33)		10%	WB	2.6	4.6	10.3	14.6	19.8	23.7	25.6	24.7	22.2	15.7	9.2	3.9	(33)
(34)			MCD B	5.1	7.4	15.2	19.3	24.4	28.5	29.4	28.8	26.2	19.9	13.2	6.4	(34)
(35)	Mean Daily Temperature Range	MDBR	8.9	9.2	10.8	11.4	10.8	10.5	10.0	10.3	11.6	11.1	9.9	8.3	(35)	
(36)		5% DB	MCD BR	12.5	13.6	15.4	15.7	13.3	11.6	10.9	12.0	13.4	14.1	13.9	12.5	(36)
(37)			MCWBR	9.1	9.1	8.9	8.5	6.7	6.1	6.6	6.8	7.0	7.8	8.6	9.5	(37)
(38)		5% WB	MCD BR	11.8	12.7	13.6	13.7	11.9	10.6	10.1	10.8	11.7	11.8	13.0	11.6	(38)
(39)			MCWBR	8.8	8.9	8.5	8.4	6.9	6.8	6.9	7.1	7.0	8.0	9.1	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.313	0.327	0.378	0.402	0.406	0.420	0.408	0.370	0.331	0.296	0.282	(40)	
(41)		taud	2.455	2.375	2.402	2.279	2.281	2.321	2.275	2.319	2.383	2.481	2.505	2.457	(41)	
(42)		Ebn at Noon	886	896	924	892	876	869	852	853	865	867	858	848	(42)	
(43)		Edh at Noon	76	96	105	128	131	127	131	122	107	86	72	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.00	2.94	3.81	4.86	5.79	6.42	6.48	5.60	4.63	3.23	2.24	1.68	(44)	
(45)		RadStd	0.18	0.18	0.35	0.34	0.44	0.51	0.43	0.36	0.30	0.38	0.16	0.17	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page